STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 03/04/2019*

EXERCISE

We observe N samples of the following discrete-time real random process:

$$X[n] = A + W[n], \quad n = 0, 1, \cdots, N-1,$$

where $A \neq 0$ is the amplitude of the Signal of Interest (SOI) and it is modelled as an unknown deterministic parameter, $W[n]$ is additive white Gaussian noise (AWGN) with Power Spectral Density (PSD) $S_W\left(e^{j2\pi f}\right) = A^2$, i.e. its power is equal to the power of the SOI (it is a signal-dependent noise).

(1) Derive the Autocorrelation Function (ACF) and the Power Spectral Density (PSD) of the w.s.s. process $X[n]$. Then, derive the joint probability density function (pdf) of the observed data vector $\mathbf{X} \triangleq \left[X[0] \quad X[1] \quad \cdots \quad X[N-1]\right]^T$ and the log-likelihood function $\ln L(A) = \ln f_X(\mathbf{x}; A)$.

(2) Derive the Method of Moments (MM) estimator of the parameter A, its bias and its Mean Square Error (MSE), and the pdf of the estimation error $\varepsilon \triangleq A - \hat{A}_{MM}$. Establish whether the estimator is biased and consistent.

(3) Derive the Maximum Likelihood (ML) estimator of the parameter A and establish whether the estimator consistent.

(4) Derive the Cramér-Rao lower bound (CRB) on the estimate of A and establish whether the Method of Moments (MM) estimator is efficient or not.

Solution

(1) ACF:

$$R_X[m] = E\{X[n]X[n+m]\} = E\{(A+W[n])(A+W[n+m])\}$$
$$= A^2 + E\{W[n]W[n+m]\} = A^2 + A^2\delta[m]$$

PSD:

$$S_X\left(e^{j2\pi f}\right) = FT\{R_X[m]\} = FT\{A^2 + A^2\delta[m]\}$$
$$= 2\pi A^2 \delta\left(e^{j2\pi f} - 1\right) + A^2$$

PDF:

$$\eta_X = E\{X[n]\} = E\{A + W[n]\} = A$$

$$\sigma_X^2 = \mathrm{var}\{X[n]\} = \mathrm{var}\{W[n]\} = A^2$$

Let's define $X_n \triangleq X[n]$. The pdf of the observed data vector is given by:

$$X_n \in N\left(A, A^2\right), \quad \{X_n\}_{n=0}^{N-1} \text{ IID } \Rightarrow \quad \mathbf{X} \in N\left(A\mathbf{1}, A^2\mathbf{I}\right)$$

The pdf and the log-likelihood function in scalar format are:

$$L(A) = f_X(\mathbf{x}; A) = \prod_{n=0}^{N-1} f_{X_n}(x_n; A) = \left(\frac{1}{\sqrt{2\pi A^2}}\right)^N \exp\left(-\frac{1}{2A^2}\sum_{n=0}^{N-1}\left(x_n - A\right)^2\right)$$

$$\ln L(A) = \ln f_X(\mathbf{x}; A) = -\frac{N}{2}\ln(2\pi) - -\frac{N}{2}\ln\left(A^2\right) - \frac{1}{2A^2}\sum_{n=0}^{N-1}\left(x_n - A\right)^2$$

(2) MM estimator of A.

Since $E\{X[n]\} = A$, we can estimate A by using the Sample Mean estimator (in this case the MM estimator coincides with the Sample Mean estimator):

$\hat{A}_{MM} = \frac{1}{N}\sum_{n=0}^{N-1} X[n]$, the MM estimator is **linear**, so it is **Gaussian distributed**.

$E\left\{\hat{A}_{MM}\right\} = E\left\{\frac{1}{N}\sum_{n=0}^{N-1} X[n]\right\} = \frac{1}{N}\sum_{n=0}^{N-1} E\{X[n]\} = \frac{1}{N}\sum_{n=0}^{N-1} A = A$, so the MM estimator is **unbiased**.

$$MSE\left\{\hat{A}_{MM}\right\} = \text{var}\left\{\hat{A}_{MM}\right\} = \text{var}\left\{\frac{1}{N}\sum_{n=0}^{N-1} X[n]\right\} = \frac{1}{N^2}\text{var}\left\{\sum_{n=0}^{N-1} X[n]\right\} = \frac{1}{N^2}\sum_{n=0}^{N-1}\text{var}\left\{X[n]\right\} = \frac{1}{N^2}\sum_{n=0}^{N-1} A^2 = \frac{A^2}{N}$$

so the MM estimator is **consistent**. The estimation error is Gaussian distributed: $\varepsilon \in N\left(0, \frac{A^2}{N}\right)$.

(3) The log-likelihood function is given by:

$$\ln L(A) = \ln f_X(\mathbf{x}; A) = -\frac{N}{2}\ln(2\pi) - -\frac{N}{2}\ln\left(A^2\right) - \frac{1}{2A^2}\sum_{n=0}^{N-1}\left(x_n - A\right)^2 .$$

Then:

$$\begin{aligned}
\frac{d\ln L(A)}{dA} &= -\frac{1}{2}\left[\frac{2N}{A} - \frac{2}{A^3}\sum\nolimits_{n=0}^{N-1}(x_n - A)^2 - \frac{2}{A^2}\sum\nolimits_{n=1}^{N}(x_n - A)\right] \\
&= -\frac{N}{A^3}\left[A^2 - \frac{1}{N}\sum\nolimits_{n=0}^{N-1}(x_n - A)^2 - A\cdot\frac{1}{N}\sum\nolimits_{n=1}^{N}(x_n - A)\right] \\
&= -\frac{N}{A^3}\left[A^2 - \frac{1}{N}\sum\nolimits_{n=0}^{N-1}(x_n^2 - 2Ax_n + A^2) - A\cdot\frac{1}{N}\sum\nolimits_{n=1}^{N}(x_n - A)\right] \\
&= -\frac{N}{A^3}\left[A^2 - \frac{1}{N}\sum\nolimits_{n=0}^{N-1}x_n^2 + 2A\frac{1}{N}\sum\nolimits_{n=0}^{N-1}x_n - A^2 - A\cdot\frac{1}{N}\sum\nolimits_{n=1}^{N}x_n + A^2\right] \\
&= -\frac{N}{A^3}\left[A^2 + A\frac{1}{N}\sum\nolimits_{n=0}^{N-1}x_n - \frac{1}{N}\sum\nolimits_{n=0}^{N-1}x_n^2\right] \\
&= -\frac{N}{A^3}\left[A^2 + S_N A - P_N\right]
\end{aligned}$$

where we defined

$$S_N \triangleq \frac{1}{N}\sum\nolimits_{n=0}^{N-1}x_n \quad \text{and} \quad P_N \triangleq \frac{1}{N}\sum\nolimits_{n=0}^{N-1}x_n^2$$

Then, the ML estimator of A is given by the solution of the equation:

$$\frac{d\ln L(A)}{dA} = -\frac{N}{A^3}\left[A^2 + S_N A - P_N\right] = 0$$

There are two solutions, only the one with sign "+" corresponds to the maximum of the log-likelihood function:

$$A = \frac{-S_N \pm \sqrt{S_N^2 + 4P_N}}{2} \qquad \Rightarrow \qquad \hat{A}_{ML} = \frac{\sqrt{S_N^2 + 4P_N} - S_N}{2}$$

The other solution (the one with sign "-") does not correspond to the global maximum.

$$\hat{A}_{ML} = \frac{\sqrt{S_N^2 + 4P_N} - S_N}{2} = \frac{\sqrt{\left(\frac{1}{N}\sum_{n=0}^{N-1} x_n\right)^2 + 4\left(\frac{1}{N}\sum_{n=0}^{N-1} x_n^2\right)} - \frac{1}{N}\sum_{n=0}^{N-1} x_n}{2}$$

$$\lim_{N\to\infty} \hat{A}_{ML} = \lim_{N\to\infty}\left(\frac{\sqrt{S_N^2 + 4P_N} - S_N}{2}\right) = \frac{\sqrt{\left(\lim_{N\to\infty} S_N\right)^2 + 4\lim_{N\to\infty} P_N} - \lim_{N\to\infty} S_N}{2} = \frac{\sqrt{A^2 + 4\left(2A^2\right)} - A}{2}$$

$$= \frac{\sqrt{9A^2} - A}{2} = \frac{3A - A}{2} = A$$

Hence, the ML estimator is **consistent**, as we expected. To derive this results we exploited the fact that the Sample Mean and Sample Power are consistent estimators:

$$\lim_{N\to\infty} S_N = \lim_{N\to\infty} \frac{1}{N}\sum_{n=0}^{N-1} x_n \; = E\{x_n\} = A$$

$$\lim_{N\to\infty} P_N = \lim_{N\to\infty} \frac{1}{N}\sum_{n=0}^{N-1} x_n^2 = E\{x_n^2\} = \left(E\{x_n\}\right)^2 + \mathrm{var}\{x_n\} = A^2 + A^2 = 2A^2$$

(4) To derive the CRB we need to calculate the second order derivative of $\ln L(A)$:

$$\frac{d^2 \ln L(A)}{dA^2} = \frac{d}{dA}\left(-\frac{N}{A^3}\left[A^2 + S_N A - P_N\right]\right) = \frac{3N}{A^4}\left[A^2 + S_N A - P_N\right] + -\frac{N}{A^3}\left[2A + S_N\right]$$

$$= \frac{N}{A^4}\left[3A^2 + 3S_N A - 3P_N - 2A^2 - AS_N\right] = \frac{N}{A^4}\left[A^2 + 2S_N A - 3P_N\right]$$

The Fisher Information is then given by:

$$I(A) \triangleq -E\left\{\frac{d^2 \ln L(A)}{dA^2}\right\} = -\frac{N}{A^4}\left[A^2 + 2E\{S_N\}A - 3E\{P_N\}\right]$$

$$= -\frac{N}{A^4}\left[A^2 + 2A \cdot A - 3 \cdot 2A^2\right] = \frac{3N}{A^2}$$

Consequently, the CRB is given by: $CRB(A)=\frac{1}{I(A)}=\frac{A^2}{3N}$.

Hence the MM estimator is not efficient, since $MSE\left\{\hat{A}_{MM}\right\}=\frac{A^2}{N}>\frac{A^2}{3N}$.

It is not even asymptotically efficient since $\frac{MSE\left\{\hat{A}_{MM}\right\}}{CRB(A)}=3 \quad \forall N$.